Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (IT) Examination

Mar 2018

IT409 High Performance Computer Architecture

Date: 28.03.2018, Wednesday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Answer the following.

a. What is set associative cache memory? Explain using suitable example. **[05]**

b. Differentiate temporal locality and spatial locality. **[02]**

Q – 2

a. List and briefly explain the policies to write data in the cache. **[04]**

b. Explain Instruction per cycle and Cycle Per Instruction using suitable example. How do they affect each other? **[04]**

c. What is direct mapped cache? Explain using suitable example. Write advantages and disadvantages of it. **[06]**

OR

Q – 2

a. Explain the LRU, LFU, FIFO and Random block replacement policies. **[04]**

b. Explain the dependencies, which may arise in the pipelining. **[04]**

c. What is fully associative mapped cache? Explain using suitable example. Write advantages and disadvantages of it. **[06]**

Q - 3

a. What is loop fusion? List the merits and demerits of the same. **[04]**

b. Write the advantages and disadvantages of Tomasulo algorithm **[04]**

c. What is an Average Memory Access Time (AMAT)? Explain using suitable example. **[06]**

OR

Q - 3

a. Explain VLIW architecture with necessary diagrams. **[04]**

b. Explain hit rate, miss rate and miss penalty. **[04]**

c. Explain the scoreboard technique using suitable example. **[06]**

SECTION – II

Q - 4

a. Explain the pipeline in brief. **[02]**

b. What is virtual memory? List and explain the three issues related to program execution, which can be solved by virtual memory. **[05]**

Q – 5.a What is TLB? Explain the working of it. Also, explain the structure of page table. **[07]**

Q – 5.b Explain the different type of memory available in the computer. **[07]**

OR

Q – 5.a Explain page fault. List the steps to recover from the page fault. **[07]**

Q – 5.b Explain the types of instruction in MIPS. **[07]**

Q – 6.a Explain distributed computing with its advantages and disadvantages. **[07]**

Q – 6.b List and explain the possible ways to reduce the miss rate in computer. **[07]**

OR

Q – 6.a Explain cluster computing with its advantages and disadvantages. **[07]**

Q – 6.b What is cache coherence? Does it applicable to single processor environment? What are the ways to solve it? **[07]**

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